

KARLSRUHE BADEN BADEN, Germany

WMO: 107275

Lat: 48.779N

Lon: 8.081E

Elev: 408

StdP: 14.48

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	19.1	22.7	11.9	10.2	23.1	17.3	13.4	26.1	27.3	48.4	24.7	41.6	6.4	30	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	90.0	69.0	86.2	68.2	82.7	66.6	71.0	84.7	69.7	82.4	68.2	79.6	6.9	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.6	99.3	75.3	65.0	93.9	74.0	64.2	91.4	73.4	35.2	84.8	34.0	82.7	32.8	79.7	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	19.6	17.3	13.1	97.0	6.5	2.5	8.4	98.8	4.6	100.2	0.9	101.6	-3.8	103.4
			12.6	74.9	6.5	1.7	7.9	76.1	4.1	77.1	0.5	78.1	-4.2	79.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.5	36.5	37.4	44.4	52.5	59.3	66.3	69.6	68.0	60.9	52.4	43.5	38.3	(6)
	DBStd	13.58	7.87	7.87	7.33	6.67	6.55	6.24	5.69	5.89	5.51	6.64	6.65	7.18	(7)
	HDD50	1668	423	355	199	50	6	0	0	0	1	53	215	366	(8)
	HDD65	4994	885	772	640	375	199	58	21	34	147	390	645	828	(9)
	CDD50	2583	3	3	24	126	293	488	607	559	328	129	20	3	(10)
	CDD65	436	0	0	0	1	22	97	163	128	24	1	0	0	(11)
	CDH74	4161	0	0	0	54	261	916	1533	1155	231	11	0	0	(12)
	CDH80	1401	0	0	0	5	52	296	585	418	45	0	0	0	(13)
(14) Wind	WSAvg	7.3	9.0	8.4	8.5	7.1	7.2	6.6	6.6	6.0	6.3	6.7	7.2	8.4	(14)
(15) Precipitation	PrecAvg	36.7	3.0	2.7	2.7	2.4	3.4	3.3	3.5	3.0	2.7	3.2	3.3	3.5	(15)
	PrecMax	46.5	6.3	5.9	6.9	4.7	7.4	7.1	7.9	6.0	5.1	8.8	6.6	8.4	(16)
	PrecMin	26.6	0.4	0.7	0.6	0.1	1.0	1.1	1.2	0.4	1.0	0.4	0.2	0.5	(17)
	PrecStd	5.4	1.4	1.5	1.7	1.2	1.5	1.4	1.5	1.3	1.1	1.7	1.5	1.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	59.3	69.6	79.2	86.1	92.8	95.3	95.4	84.5	74.9	64.2	57.4	(19)
		MCWB	50.6	48.8	53.3	59.1	66.8	69.7	70.4	70.1	65.5	63.6	56.5	51.2	(20)
	2%	DB	54.0	55.2	64.7	73.8	80.6	87.7	91.0	89.3	80.4	69.6	60.0	54.0	(21)
		MCWB	49.0	47.7	51.2	57.0	63.5	68.2	69.2	69.1	65.9	61.0	54.0	50.0	(22)
	5%	DB	51.8	51.7	60.8	70.0	76.8	83.9	86.3	84.4	75.5	65.9	57.1	51.7	(23)
		MCWB	47.2	45.9	49.7	55.9	61.4	67.5	67.9	67.9	63.8	58.7	52.3	47.8	(24)
	10%	DB	48.5	48.4	56.9	66.2	72.7	79.2	82.4	80.5	71.7	62.5	53.8	48.5	(25)
		MCWB	45.1	44.1	48.8	53.9	59.9	65.4	66.2	66.6	61.8	57.0	49.8	45.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.3	50.8	55.1	61.2	69.6	73.2	72.6	73.0	69.1	63.9	57.6	53.3	(27)
		MCDB	56.2	56.6	65.4	75.5	81.7	86.3	88.3	86.3	79.9	71.3	62.1	56.0	(28)
	2%	WB	50.1	48.6	53.1	58.5	66.5	70.2	70.7	70.9	66.9	61.5	54.6	50.7	(29)
		MCDB	53.7	54.1	60.7	71.0	77.4	83.2	85.0	84.0	76.8	67.4	58.6	53.8	(30)
	5%	WB	47.6	46.4	51.2	56.5	63.5	68.5	69.4	69.3	64.8	59.6	52.6	48.1	(31)
		MCDB	51.4	50.8	58.5	67.8	72.7	80.3	82.3	81.5	73.5	64.7	56.7	51.3	(32)
	10%	WB	45.1	44.0	49.3	54.7	61.1	66.8	67.8	67.5	62.8	57.7	50.4	45.8	(33)
		MCDB	47.9	47.7	56.1	64.2	69.6	78.0	79.1	78.0	70.2	61.7	53.5	48.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	12.3	17.4	19.8	19.0	19.5	19.9	19.9	19.1	15.5	9.9	9.0	(35)
		MCDBR	12.4	17.8	26.0	28.8	26.9	25.9	28.2	26.9	25.3	20.4	14.3	12.0	(36)
	5% WB	MCWBR	9.3	11.7	14.7	14.1	12.1	10.9	10.1	10.6	12.5	12.4	9.5	9.3	(37)
		MCWBR	11.1	15.7	21.1	25.2	22.6	23.6	23.9	24.7	22.1	18.6	13.2	11.5	(38)
(40) Clear-Sky Solar Irradiance	taub		0.322	0.347	0.379	0.412	0.422	0.427	0.424	0.421	0.394	0.373	0.342	0.319	(40)
		taud	2.424	2.355	2.286	2.225	2.237	2.260	2.257	2.290	2.336	2.385	2.418	2.415	(41)
	Ebn at Noon		234	252	262	265	267	266	264	259	254	239	222	218	(42)
		Edh at Noon	21	28	35	41	42	42	41	38	33	27	21	19	(43)
(44) All-Sky Solar Radiation	RadAvg		291	531	912	1354	1600	1823	1708	1485	1111	665	336	232	(44)
	RadStd		42	90	136	187	174	149	162	147	121	63	39	38	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (72 neighbors)	N/A	N/A	N/A	N/A	+0.61	+0.49	N/A	N/A	+92	N/A	

Nomenclature: See separate page